

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0608

Main 138 kV
January 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0608. The Interconnection Customer proposes a 143.1 MW solar photovoltaic facility interconnecting at the Main 138 kV substation in the Delaware area. The request was received on April 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,120,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Keney Substation – Dell Crk 138 kV	Red Lion Energy Center – Delaware City Plant 138 kV	127.9	196	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dell Creek – Keeney 138 kV	Rebuild line with higher-capacity conductor	\$2,120,000	100% (\$2,120,000)
	Total allocated to project		\$2,120,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.